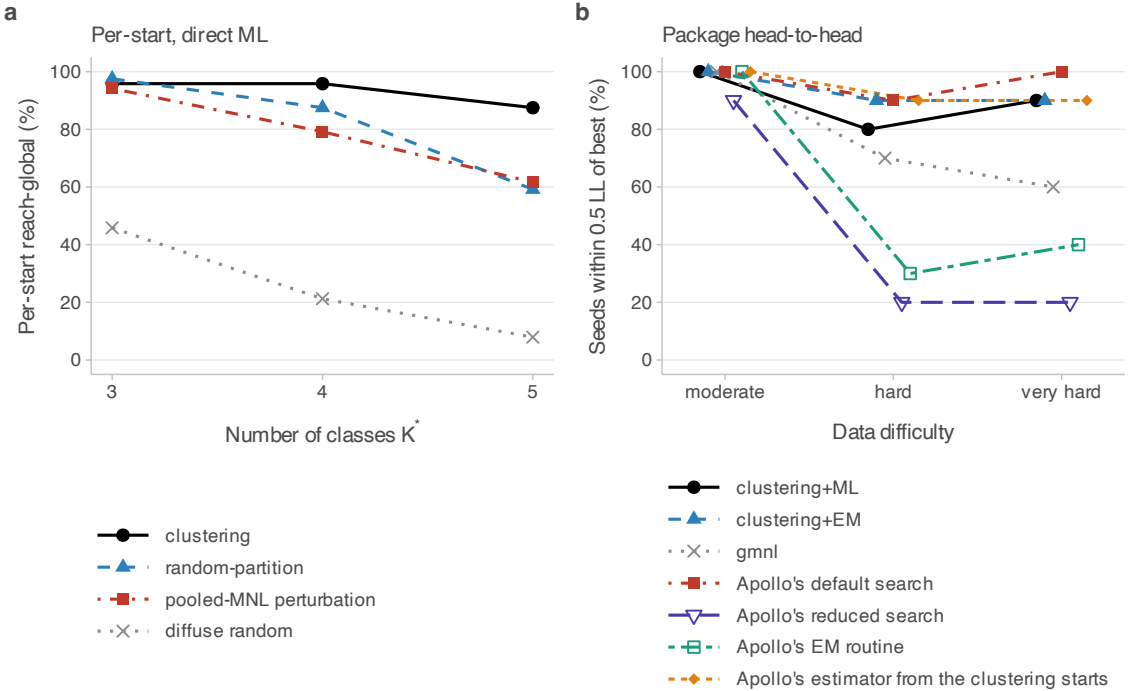
a
Per-start, direct ML
Per-start reach-global (%)
100
80
60
40
20
0
3
4
5
Number of classes K*
clustering
random-partition
pooled-MNL perturbation
diffuse random
b
Package head-to-head
Seeds within 0.5 LL of best (%)
100
80
60
40
20
0
moderate
hard
very hard
Data difficulty
clustering+ML
clustering+EM
gmnl
Apollo's default search
Apollo's reduced search
Apollo's EM routine
Apollo's estimator from the clustering starts